

15.6 Using the NLS panel data on $N = 716$ young women, we consider only years 1987 and 1988. We are interested in the relationship between $\ln(WAGE)$ and experience, its square, and indicator variables for living in the south and union membership. Some estimation results are in Table 15.10.

TABLE 15.10 Estimation Results for Exercise 15.6

	(1) OLS 1987	(2) OLS 1988	(3) FE	(4) FE Robust	(5) RE
<i>C</i>	0.9348 (0.2010)	0.8993 (0.2407)	1.5468 (0.2522)	1.5468 (0.2688)	1.1497 (0.1597)
<i>EXPER</i>	0.1270 (0.0295)	0.1265 (0.0323)	0.0575 (0.0330)	0.0575 (0.0328)	0.0986 (0.0220)
<i>EXPER</i> ²	−0.0033 (0.0011)	−0.0031 (0.0011)	−0.0012 (0.0011)	−0.0012 (0.0011)	−0.0023 (0.0007)
<i>SOUTH</i>	−0.2128 (0.0338)	−0.2384 (0.0344)	−0.3261 (0.1258)	−0.3261 (0.2495)	−0.2326 (0.0317)
<i>UNION</i>	0.1445 (0.0382)	0.1102 (0.0387)	0.0822 (0.0312)	0.0822 (0.0367)	0.1027 (0.0245)
<i>N</i>	716	716	1432	1432	1432

(standard errors in parentheses)

- The OLS estimates of the $\ln(WAGE)$ model for each of the years 1987 and 1988 are reported in columns (1) and (2). How do the results compare? For these individual year estimations, what are you assuming about the regression parameter values across individuals (heterogeneity)?
- The $\ln(WAGE)$ equation specified as a panel data regression model is

$$\ln(WAGE_{it}) = \beta_1 + \beta_2 EXPER_{it} + \beta_3 EXPER_{it}^2 + \beta_4 SOUTH_{it} + \beta_5 UNION_{it} + (u_i + e_{it}) \quad (XR15.6)$$

- Explain any differences in assumptions between this model and the models in part (a).
- Column (3) contains the estimated fixed effects model specified in part (b). Compare these estimates with the OLS estimates. Which coefficients, apart from the intercepts, show the most difference?
- The F -statistic for the null hypothesis that there are no individual differences, equation (15.20), is 11.68. What are the degrees of freedom of the F -distribution if the null hypothesis (15.19) is true? What is the 1% level of significance critical value for the test? What do you conclude about the null hypothesis.
- Column (4) contains the fixed effects estimates with cluster-robust standard errors. In the context of this sample, explain the different assumptions you are making when you estimate with and without cluster-robust standard errors. Compare the standard errors with those in column (3). Which ones are substantially different? Are the robust ones larger or smaller?
- Column (5) contains the random effects estimates. Which coefficients, apart from the intercepts, show the most difference from the fixed effects estimates? Use the Hausman test statistic (15.36) to test whether there are significant differences between the random effects estimates and the fixed effects estimates in column (3) (Why that one?). Based on the test results, is random effects estimation in this model appropriate?

(a-e)

請見前一次作業

(f)

EXPER 在 FE 和 RE 之間差異最大!

FE 係數: 0.0575 $\rightarrow$ RE 係數: 0.0982

因 EXPER 和 μ_i (例: 能力) 高度正相關, RE 忽略了這個內生性, 導致 EXPER 的係數被高估。

Hausman test:

$$t = \frac{b_{FE} - b_{RE}}{\sqrt{[se(b_{FE})]^2 - [se(b_{RE})]^2}}$$

$$\rightarrow t^* = \frac{0.0575 - 0.0982}{\sqrt{0.033^2 - 0.022^2}} \approx -1.65$$

$$PR = \{t: |t| > t_{0.025}(n) = 1.96\}$$

$$\rightarrow NT - N - K_5 = 1432 - 1716 - 4 = -912$$

$$t^* \notin PR$$

$\rightarrow$ do not reject H_0 , 無法拒絕「固定效果模型和隨機效果模型的係數無顯著差異」的假設

$\rightarrow$ 採用 random effects estimation 是合適的。

補: why that one?

因 EXPER FE 和 RE 算出來的係數產生了高達近一倍的差距 (0.0575 vs 0.0982), 最有可能受到遺漏變數 (例如: 能力) 的內生性衝擊。之所選 EXPER 來做檢定, 是為了檢驗模型中是否存在因為『遺漏個人能力』所導致的隨機效果偏誤。